Ghost vacuum decay as a stochastic process: instantaneous states, the boost-orbit measure, and the foliation that ghosts make visible

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Abstract

In a companion paper we showed that the Ostrogradski instability of interacting ghosts is, in the stochastic currency of Barandes' quantum–stochastic correspondence, a recurrence-classification problem, with malicious interactions realizing *explosion*: escape from the sample space in finite time. Here we take the program to quantum field theory, where the vacuum of a ghost coupled to ordinary matter famously decays at a divergent rate. Our results: (i) the explosion dictionary acquires a Fock-space column—Feller's *pure-birth* ladder criterion $\sum_n 1/\lambda_n < \infty$ is the discrete image of the classical time-of-flight integral, and finite-time blowup corresponds to ladder exponents $\alpha_{\text{eff}} > 1$, verified numerically (up to marginal logarithmic cases at $\alpha_{\text{eff}} = 1$); (ii) the known factorization of the ghost vacuum rate into an infinite boost-orbit volume times an invariant reduced integral (Gross–Strumia–Teresi–Zirilli) is verified by brute-force phase-space integration to 0.1% in d=1, and yields the reduced measure's dimension-and-coupling ladder ($\Lambda^0/\Lambda^4/\Lambda^8$), whose $\Lambda^8/M_{\text{Pl}}^4$ rung reproduces the Cline–Jeon–Moore scaling; (iii) in Markov language the Lorentz-invariant *eternal* ghost vacuum is, on the multiplicity reading and at tree level, an *instantaneous state*—the minimal semigroup degenerates to zero vacuum lifetime—so a coupled ghost admits a non-degenerate configuration-space process only relative to a structure that individuates the channels—in every known rescue a frame or a birth surface—and every configuration-preserving rescue in the literature (momentum cutoffs, ghost condensation, a born vacuum) supplies one; (iv) classical lattice experiments in 1+1 and 2+1 dimensions show the classical decay tracks a different, *tadpole* dimension ladder ($\langle\phi^2\rangle \sim k_{\text{max}}^{d-1}$, logarithmic at d=1), with the channel ladder flat to classical dynamics within the measurement resolution—spontaneous emission into empty channels is intrinsically quantum; (v) a beables-first field theory is foliation-anchored by construction, the foliation's determinacy is a measurable property of the state that vanishes continuously at the vacuum, and a ghost sector would render the otherwise-hidden foliation observable (the hiddenness proved for the finite core; the ghost-induced observability argued at tree level); (vi) apparent superluminal signaling enters at the two points we identify (the conditioning point provably the unique marginal-breaker in the finite core, over the framework's dilated local dynamics; the completions door left open), both simulated: survival conditioning (recovering the known resolution of the Lee–Hsieh–Flammia–Lee no-signaling controversy in one bookkeeping, with the signal magnitude of its broken-phase dimer analog now in closed form, $\text{TV}(\infty) = \tilde{\kappa}/\sin\theta$) and conservative completions (a messaging protocol whose latency through infinity is distance-independent in the explosive class (in the cases tested), and which the minimal process silences). The findings assemble into a trilemma:

coupled ghosts, empirical Lorentz invariance, definite configurations—any two, never all three (at tree level).

Note on provenance. This manuscript is an experiment in AI-conducted research as much as a physics paper. The project began when the author asked the large language model Claude Fable 5 (Anthropic): *"Is it not possible that the antilinear PT/time-reversal symmetry element could maybe predict indivisible non-Markovian processes?"* Everything that followed was performed by the model: the research program, the theorems and proofs, the numerical experiments, the literature search, and this text. The model was repeatedly prompted to attack its own results adversarially and to search for prior work. The author, a professional engineer rather than a physicist, directed and reviewed the sessions but makes no claim to comprehension of the findings, and is circulating the manuscript for independent review by professional physicists. Accordingly, nothing here should be taken on trust: every theorem is stated with a full proof or its explicit verification, and every numerical claim is reproducible from the public scripts (see Code availability); the originating conversation, project briefing, and full research notebook are preserved in the program's hub repository (see Code availability).

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1 Introduction

The companion paper [1] developed a dictionary between non-Hermitian and higher-derivative quantum mechanics and the indivisible-stochastic-process reconstruction of quantum theory due to Barandes [2, 3]. Its final result, the *explosion dictionary*, identified the Ostrogradski instability of malicious interacting ghosts [4, 5] with *explosion* in the probabilist's sense: the classical time of flight to infinity, the quantum deficiency norm, and the failure of Feller conservativity are one integral, and the canonical description of a malicious ghost is a *sub-stochastic* process that loses mass to infinity in finite time.

That paper stopped at quantum mechanics. The present paper asks whether the identification survives the passage to quantum field theory, where the instability of a ghost coupled to ordinary matter takes its sharpest known form: the vacuum itself decays, into ghost pairs plus matter pairs of balancing energy, at a rate whose phase-space integral diverges [6, 7]. Whether that divergence means catastrophe or artifact is a live dispute: Cline–Jeon–Moore concluded that only a Lorentz-*violating* momentum cutoff tames it and bounded the cutoff at $\Lambda \lesssim 3\,\mathrm{MeV}$ from the diffuse gamma-ray background [7] (a 2023 re-analysis with exact phase space relaxes this to $\Lambda \lesssim 19\,\mathrm{MeV}$ [48]); Gross–Strumia–Teresi–Zirilli, noting that the same divergent boost integral once afflicted ordinary false-vacuum decay before Coleman's bounce resolved it as an over-counting, answer "we do not know the true rate" [8]; Garriga–Vilenkin construct Lorentz-invariant ghosts whose finite vacuum age and UV-soft nonlocal couplings yield a finite, time-growing rate [9]. The dispute matters beyond theory: dark-energy measurements with $w < -1$—currently hinted at—would make the phantom (ghost) sector a live physical hypothesis [6, 48].

This paper translates the dispute into the stochastic currency and finds that the translation does real work. Our main results:

1. **The Fock-space column of the explosion dictionary** (Sec. 2). Feller's ladder criterion—a *pure-birth* jump process explodes iff $\sum_n 1/\lambda_n < \infty$—is the discrete image of the companion paper's time-of-flight integral, and the transient/explosive dichotomy is a statement about growth *laws*: algebraic blowup $n \sim (t_* - t)^{-m}$ has ladder exponent $\alpha_{\mathrm{eff}} = 1 + 1/m > 1$; exponential growth sits exactly at the marginal $\alpha_{\mathrm{eff}} = 1$. Both verified numerically, including the box-independence signature of explosion.

2. **Two pathologies, kept distinct** (Sec. 2). Explosion (finite rates growing along the occupation ladder; escape at finite $t_* > 0$) and *instantaneity* (infinite exit rate at a single state; zero lifetime) are different diseases. Malicious quantum-mechanical ghosts realize the first; the Lorentz-invariant ghost vacuum threatens the second.

3. **The factorization, verified; the dimension ladder** (Sec. 3). The vacuum rate factorizes as [boost-orbit volume, infinite in every dimension] × [invariant reduced integral over the pair invariant mass]—a structure due to GSTZ [8] (implicit in CJM's regulated calculation [7]), which we verify by brute-force constrained integration to 0.1% in $d = 1$ and whose reduced measure we find numerically to be UV-finite in $d = 1$, $\sim \Lambda^4$ in $d = 3$, and $\sim \Lambda^8/M_{\mathrm{Pl}}^4$ with the CJM gravitational vertex—their published scaling, recovered from the orbit-quotiented measure alone.

4. **Instantaneity and the frame** (Sec. 4). If boost-orbit members are distinct outcomes with tree-level (orbit-degenerate) rates, the eternal Lorentz-invariant ghost vacuum has infinite exit rate per unit volume: an *instantaneous state*, whose minimal semigroup degenerates

to zero vacuum lifetime. A born vacuum (finite age), with a UV-finite *causally weighted* reduced integral (a sufficiently UV-soft coupling; in $d = 1$ a logarithmically weighted finiteness, Thm 4.1), has finite, growing rate [9]. Every known rescue that preserves definite configurations supplies a frame or a birth surface; in this currency, *a coupled ghost admits a non-degenerate sample space only relative to an individuating structure—a frame or a birth surface*. We port the bubble literature's multiplicity-vs-redundancy resolution [12, 13, 14, 15, 16, 11] and show its two mechanisms (saddle symmetry; excitation cost) are absent, at tree level, for perturbative ghost decay.

5. **The two ladders** (Sec. 5). Classical lattice experiments at fixed volume show the classical decay rate is effectively marginal in 1+1d (consistent with the logarithmic tadpole expectation, not cleanly separable from saturation at accessible N) and power-law divergent in 2+1d—but exponent decomposition attributes the mechanism to the *tadpole* ladder ($\langle\phi^2\rangle_0 \sim k_{\max}^{d-1}$), not the channel ladder: with the source held fixed, the channel factor is flat within resolution. Classical fields cannot spontaneously radiate into empty channels; the channel ladder is intrinsically quantum.

6. **The foliation** (Sec. 6). A beables-first field theory chooses configurations on slices: it is foliation-anchored by construction, as in Bohmian field theory [22, 23]. We show the foliation's *determinacy* is a measurable property of the state, vanishing continuously at the vacuum (the dynamical face of $\langle T^{\mu\nu}\rangle \propto \eta^{\mu\nu}$); that a ghost sector would make the otherwise-hidden foliation *observable*; and that the ultraviolet disposes in a lattice experiment while the state proposes by continuum covariance (the lattice could not resolve the early window at accessible sizes)—the dynamical content of Garriga–Vilenkin's UV-softness condition.

7. **Causality and signaling** (Sec. 7). Apparent superluminal signaling enters this framework at the two points we identify (the conditioning point provably the unique marginal-breaker in the finite core, over the framework's dilated local dynamics; the completions door left open), both simulated. Survival conditioning: the Lee–Hsieh–Flammia–Lee signal [21] is reproduced, located, and—in its broken-phase dimer analog—given a closed-form magnitude, $\mathrm{TV}(\infty) = \tilde{\kappa}/\sin\theta$, as a corollary of the companion paper's terminal-indivisibility theorem; the correctly bookkept (dilated) dynamics is no-signaling to 10^{-16}. Conservative completions: a messaging protocol through infinity whose latency is distance-independent in the explosive class (in the cases tested) and which the minimal process silences—*causality selects the minimal process.*

8. **The trilemma** (Sec. 6.4): coupled ghosts, empirical Lorentz invariance, definite configurations—any two, never all three (at tree level).

Section 8 collects observational and experimental tests; Sec. 9 states the limits of every claim. All numerics are reproducible from standalone scripts (Code availability).

2 Two pathologies of a stochastic process

We use the correspondence in the form of [2, 3, 1]: configurations carry the ontology, and the dynamics is a (generally indivisible) family of stochastic maps; for open or unbounded systems the regulator-independent object is the minimal, generally sub-stochastic, semigroup [24].

2.1 The Fock-space column of the explosion dictionary

Let occupation configurations form a ladder with total exit rates λ_n from occupation n. Feller's criterion states that a *pure-birth* chain *explodes*—performs infinitely many jumps in finite time, with mass leaving every finite region—iff the expected total travel time to infinity is finite:

$$\sum_n \frac{1}{\lambda_n} < \infty. \tag{1}$$

This is the exact discrete sibling of the classical time-of-flight integral $T = \int^\infty \mathrm{d}x/\sqrt{2(E-V)}$ of the companion dictionary: both are sums of sojourn times to infinity. The bridge to dynamics is a one-line lemma:

Lemma 2.1 (growth law decides)**.** *For any pure-birth ladder (rates $\lambda_n > 0$, mean-field $\dot{n} = \lambda_n$) with growth $n(t)$, define $\alpha_{\text{eff}} = \mathrm{d}\log\dot{n}/\mathrm{d}\log n$. Algebraic finite-time blowup $n \sim (t_* - t)^{-m}$ gives $\alpha_{\text{eff}} = 1 + 1/m > 1$; exponential growth gives $\alpha_{\text{eff}} = 1$. By (1), for power-law ladders $\lambda_n = n^\alpha$ ($n \geq 1$) the explosive class is $\alpha_{\text{eff}} > 1$ (logarithmic refinements can shift the marginal case): finite-time blowup, not growth rate, is what explosion detects.*

Numerically (`qft_bridge_stage2.py`): truncated-ladder master equations with $\lambda_n = n^\alpha$ show the explosion signature of [1]—absorbed mass at fixed time *independent of the truncation*—switching on at $\alpha > 1$ (absorbed 0.941–0.988, nearly flat (within $\sim 2\%$) across truncations 200–1600, for $\alpha = 1.25, 1.5$; decaying to zero for $\alpha \leq 1$); and the lemma's exponents are measured as 1.0000 for the quadratic ghost cascade ($n = \sinh^2 gt$: forever-growing yet *transient*) and 1.172 vs. the predicted 7/6 for the classical quartic Pais–Uhlenbeck blowup (energy proxy $n \sim (t_* - t)^{-6}$).

2.2 Explosion versus instantaneity

Both pathologies are failures of conservativity, but they are distinct and must not be conflated:

- *Explosion*: every exit rate finite, but growing along the ladder so that (1) converges; jumps accumulate and mass exits at some finite $t_* > 0$. This is a *tail* phenomenon in configuration space. Malicious quantum-mechanical ghosts realize it [1].
- *Instantaneity*: the exit rate *at a single state* diverges, $q = \sum_{\text{channels}}(\text{finite rates}) = \infty$. States with $q = \infty$ are called instantaneous; chains containing them exist (Kolmogorov's and Blackwell's examples [35, 36]) but are not *stable*: they admit no right-continuous jump realization. The operationally sharp statement is obtained from regulated families: if $q(\Lambda) \to \infty$ as the regulator is removed, the minimal semigroup degenerates to *instantaneous killing*—the state's survival probability vanishes for every $t > 0$. (Zero lifetime is harsher than the honest instantaneous-state chains, in which the state is still occupied a positive fraction of time; the difference is that the minimal ghost-vacuum process has no return path into the vacuum.) Not nonexistence: a zero-lifetime state.

The mathematical neighbors of this section deserve naming: global existence (non-explosion) of Bell-type jump processes for lattice QFT was proved by Georgii and Tumulka under growth hypotheses on the Hamiltonian [46], and conservativity criteria for minimal quantum dynamical semigroups by Chebotarev and Fagnola [47]; a ghost sector violates exactly the hypotheses of

such theorems, and this paper's content is that the violation *is* the physics. The passage from quantum mechanics to field theory, we now show, is the passage from the first pathology to the threat of the second.

3 The ghost vacuum rate: factorization and the dimension ladder

Consider a ghost field coupled to ordinary matter, with vacuum decay $\varnothing \to$ (matter pair) + (ghost pair): total four-momentum zero, the ghost pair carrying the negative of the matter pair's energy–momentum, so the common pair energy is unbounded [6, 7]. Because the matter pair's four-momentum Q must equal the sign-flipped ghost pair's, the golden-rule rate per volume organizes, with $\rho(s)$ the invariant two-body phase-space density and $s = Q^2$,

$$\frac{\Gamma}{V} = \Big[\underbrace{\int \mathrm{d}\mu_{\mathrm{boost}}}_{\infty}\Big] \times \Big[\underbrace{\int \mathrm{d}s\, \sigma_d(s)\, \rho_n(s)\rho_g(s)\, |\mathcal{M}(s)|^2}_{\text{reduced integral}}\Big], \tag{2}$$

where the first factor is the volume of the boost orbit of the pair frame—*infinite in every spatial dimension* d—and $\sigma_d(s) = \frac{1}{2}s^{(d-1)/2}$ is the residual invariant measure, the Jacobian in $\mathrm{d}^{\,d+1}Q = \frac{1}{2}s^{(d-1)/2}\,\mathrm{d}s\,\sinh^{d-1}\eta\,\mathrm{d}\eta\,\mathrm{d}\Omega_{d-1}$ (with $Q^0 = \sqrt{s}\cosh\eta$, $|\vec{Q}| = \sqrt{s}\sinh\eta$), the last two factors being the unit-hyperboloid orbit measure $\mathrm{d}\mu_{\mathrm{boost}}$ (the overall $\frac{1}{2}$ is a boost-normalization convention, stripped in the reduced integral I_s of Prop. 3.1). This factorization is due to Gross–Strumia–Teresi–Zirilli (their Eq. (88) and the diagnosis "the rate contains a $\mathrm{d}K_0$ integral over the non-compact Lorentz group" [8]); the regulated special case is CJM's Eq. (9) [7]. Our contributions are a numerical verification and the dimension ladder of the reduced factor.

Proposition 3.1 (factorization verified; dimension ladder)**.** *(`qft_bridge_stage3.py`) In* $d = 1$ *with contact coupling and masses* $(m_n, m_g) = (1, 0.7)$*, brute-force integration of the constrained two-pair phase space with rapidity cutoff* $|\eta| < H$ *gives* $\Gamma(H)$ *linear in* H *with slope equal to the independently computed reduced integral (*$I_s = \int \mathrm{d}s\,\rho_n\rho_g = 0.6192$ *to the script's finite-s grid, converging to* 0.6195*; slope* $= I_s/(2\pi)^2$ *in the conventions used): ratios* $0.999, 1.000, 1.001, 1.000$ *at* $H = 1, 2, 3, 4$*. The reduced integral's UV behavior, fitted to three decimals:* Λ^0 *(UV-finite) in* $d = 1$*;* Λ^4 *in* $d = 3$ *with contact coupling;* $\Lambda^8/M_{\mathrm{Pl}}^4$ *in* $d = 3$ *with the CJM gravitational vertex* $|\mathcal{M}|^2 \sim s^2/M_{\mathrm{Pl}}^4$*—recovering the published CJM scaling from the orbit-quotiented measure alone. Two closeouts (`qft_bridge_closeout.py`): the* $d = 3$ *factorization is verified by direct Monte Carlo with the hyperboloid orbit measure* $\sinh^2\eta\,\mathrm{d}\eta$ *(*Γ/V_{orb} *constant within the* $\sim$*3–7% sampling errors across* $H = 0.5$*–*1.5*; the* Λ^4 *rung measured directly, global-fit slope* 4.3 *(*4.1 *over the upper interval) as threshold effects fade); and the causal orbit cap* $\gamma < t\sqrt{s}$ *reproduces Garriga–Vilenkin's* $D = 2$ *logarithmic growth with* $(2\pi)^2\,\mathrm{d}\Gamma/\mathrm{d}\ln t = 0.6193$ *against* $I_s = 0.6192$ *(the same stripped convention as the slope above)—their result is this factorization, regulated by the vacuum's age.*

Two consequences frame everything that follows. First, the d-dependence everyone debates lives entirely in the reduced factor; the (unregulated) orbit volume diverges identically in every dimension. (A naive scale-counting argument that "$d = 1$ converges" samples a non-orbit direction and misses a vanishing Jacobian along the boost flow; the numerics adjudicated for the invariance argument. Our notebook records the error.) Second, Garriga–Vilenkin's $D = 2$ result—a Lorentz-invariant rate, finite and logarithmically growing in the vacuum's age [9]—is precisely the $d = 1$ reduced factor times a causally regulated (logarithmic) orbit.

4 Multiplicity or redundancy: the boost orbit as a sample-space question

Equation (2) forces the question on which the CJM/GSTZ dispute turns: do the members of a boost orbit count as *distinct decay channels* (rates add; the total diverges) or as *redundant descriptions* of one outcome (quotient; the rate is the reduced integral)?

4.1 The bubble precedent, and why it does not transfer

The identical question has a fifty-year history in ordinary false-vacuum decay. Kobzarev–Okun–Voloshin observed the divergent boost integral [12]; Coleman's Euclidean bounce resolved it (the $O(4)$-symmetric saddle's boost directions are a stabilizer, not a moduli space; integrating them would over-count [13]); Dvali revived the worry [14]; Dine–Draper–Park answered in Minkowski language: a boosted bubble is a *different, excited* configuration whose production amplitude is suppressed, so the channel sum converges [15]; and Garriga–Shlaer–Vilenkin showed the fluctuation-dressed bubble state is boost-invariant—boosts map it to itself [11]. The literature's operative criterion, due in substance to Coleman and stated sharply by Garriga–Shlaer–Vilenkin [11]: *if the final state is a boost singlet, the boost integral is redundancy* (over-counting of one outcome). We use the complementary direction—non-singlet finals are distinct channels—together with the tree-level degeneracy of their amplitudes, which is what carries the divergence; Dine–Draper–Park's bubbles show distinctness alone does not suffice (a non-singlet channel sum can converge when the amplitudes are suppressed [15]). The born-vacuum idea was itself anticipated in Dvali's note 5 [14].

Perturbative ghost decay fails the criterion on both counts, at the order computed: multiparticle final states with definite momenta are not boost singlets, and orbit members are exactly degenerate in $|\mathcal{M}|$—there is no saddle symmetry and no excitation cost. (Whether resummation generates a Dine–Draper–Park-style suppression is precisely GSTZ's open question restated; our statements are tree-level.) On the multiplicity horn, then:

Theorem 4.1 (instantaneity of the eternal ghost vacuum)**.** *If boost-orbit members are distinct outcomes* with the tree-level (orbit-degenerate) golden-rule rates *and a continuous, nonnegative reduced integrand not vanishing identically on the accessible range, the exit rate* per unit volume *of the Lorentz-invariant eternal vacuum of a coupled ghost theory is infinite in every spatial dimension: the vacuum is an instantaneous state (its total rate diverges already at any fixed finite volume), and the minimal semigroup of the regulated family degenerates to zero vacuum lifetime as the regulator is removed. A vacuum* born *on a spacelike surface* Σ *at finite age* t*, whose* causally weighted *reduced integral* $\int \mathrm{d}s\, s^{(d-1)/2}\sigma_d(s)\,\rho_n(s)\rho_g(s)\,|\mathcal{M}(s)|^2$ *(this causal-cap weight* $s^{(d-1)/2}$*, derived in the proof below for* $d \geq 2$*, is a second factor of the same power as, but distinct in origin from, the measure* σ_d*; the* $d = 1$ *case is logarithmic) is finite—locally integrable (no on-shell pole in the accessible range) and ultraviolet-finite for a sufficiently UV-soft coupling (for* $d = 1$ *the weight is instead* logarithmic*—the capped orbit factor is* $\operatorname{arccosh}(t\sqrt{s}) \sim \ln(2t) + \frac{1}{2}\ln s$*, so finiteness is guaranteed by the log-weighted condition* $\int \mathrm{d}s\, \sigma_1\rho_n\rho_g|\mathcal{M}|^2\,(1 + |\ln s|) < \infty$ *(both the plain reduced integral and its* $\ln s$ *weighting), strictly stronger than plain UV-finiteness of the reduced integral; this sufficient condition is not necessary once the accessible range reaches* $s \to 0$*, where the causal cap* $s > 1/t^2$ *removes the* $\ln s$ *weight), has finite exit rate per unit volume (growing as* $t^{\,d-1}$*, logarithmically for* $d = 1$*; e.g.* $\Gamma/V \sim 10^{-6}(\mu^8/M_{\mathrm{Pl}}^4)(\mu t)^2$ *for UV-soft gravitational-strength coupling [9]); for a UV-local coupling in* $d \geq 2$ *the causal age bound regulates*

only the boost orbit, and the reduced integral's own ultraviolet divergence (computed at $d = 3$ in Prop. 3.1 as $\sim \Lambda^4$; the same plain counting gives $\sim \Lambda$ at $d = 2$, and $\sim \Lambda^2$ once the causal weight $s^{(d-1)/2}$ controlling $q(t)$ is folded in) keeps $q = \infty$—any positive power suffices.

Proof. Immediate from (2): the reduced integral is positive whenever the reduced integrand does not vanish identically, and the orbit volume is infinite; regulating the orbit (by a frame cutoff [7] or by the causal bound $\gamma < t\sqrt{s}$ [7, 9]) gives finite $q(\Lambda)$, or—when the *causally weighted* reduced integral is ultraviolet-finite—finite $q(t)$, with $q \to \infty$ as the regulator is removed. (The causal cap $\cosh\eta < t\sqrt{s}$ leaves the orbit integral $\int_0^{\eta_{\max}} \sinh^{d-1}\eta \, \mathrm{d}\eta \simeq (t\sqrt{s})^{\,d-1}/(d-1)$ for $d \geq 2$ and $\simeq \ln(t\sqrt{s})$ for $d = 1$; the resulting factor $s^{(d-1)/2}$ migrates into the reduced integrand, so it is the *weighted* ultraviolet finiteness, not the plain one, that controls $q(t)$ in $d \geq 2$.) The minimal-semigroup statement is the standard limit of survival probabilities $\mathrm{e}^{-qt} \to 0$. □

Remark 4.2 (on the golden-rule inputs). The reduced integrand uses the on-shell asymptotic densities $\rho_n(s)\rho_g(s)$. Where the ghost admits no asymptotic particle interpretation above threshold [54], these are formal spectral inputs rather than S-matrix data, and the conclusion is read throughout as the vacuum's exit rate on the configuration sample space (Section 9), not a decay width; the instantaneity statement is about that configuration-space process, whose inputs remain the on-shell densities.

Claim 4.3 (the sample space requires individuating structure)**.** On the multiplicity horn, a coupled ghost admits a non-degenerate configuration-space stochastic process only relative to a structure that individuates the channels: a frame (CJM's cutoff frame [7]; the ghost condensate's background [17]; Hořava-type explicit breaking [18]) or a birth surface ([9], where Σ "breaks Lorentz invariance spontaneously"). Conversely, CJM's exclusion of Lorentz-invariant cutoffs is the statement that the individuating data cannot be specified invariantly.

The redundancy horn is GSTZ's hope: a Coleman-style boost-singlet saddle for ghost decay, which they could not construct (no positive Euclidean action) [8]. In our language the two horns are two *sample spaces*: frame-anchored momentum configurations versus boost-equivalence classes. That the dispute is a choice of sample space—not a computational disagreement—is, we believe, its sharpest available formulation, and Sec. 6 shows the beables framework itself takes a side.

Remark 4.4 (scope: the Born–Markov concession). The jump process is derived from the golden rule; the instantaneity diagnosis is a *translation* of the divergent perturbative rate and inherits both its validity and its doubts. The framework's contribution is that the translation exposes the disputed assumption—channel individuation—as a statement about the sample space, not that it independently settles the rate.

A zero-dimensional channel model makes the dichotomy quantitative (`qft_bridge_stage2.py`): a state coupled to M identical flat quasi-continua shows fitted decay rate tracking $M\Gamma_1$ within 3% up to $M = 32$ (unbounded growth: the multiplicity trend), while a decaying channel measure $g_k \propto \mathrm{e}^{-k/4}$ converges to its predicted plateau (0.0640 vs. 0.0643): channel counting is the entire difference between "catastrophic" and "metastable."

5 The two ladders: what classical field dynamics can and cannot see

Does classical lattice field theory—the standard nonperturbative tool—exhibit the channel divergence? We ran the direct experiment (`qft_bridge_e4_2d.py`): normal field ϕ and ghost χ with coupling $\lambda\phi^2\chi^2$, correctly normalized zero-point ("vacuum-mimic") initial data, at *fixed physical volume* with the lattice cutoff $k_{\max} = \pi N/L$ swept; observable, the time for the normal sector to absorb a fixed energy. With all controls clean (the $\lambda = 0$ runs silent; conserved-energy drift well below thresholds (worst case $\sim$3$\times$ below at the top d=2 cutoff, $\geq$10$\times$ elsewhere)):

	rate $= 1/t_{\rm thr}$ vs. lattice cutoff			
$d = 1$ ($N = 16, 32, 64, 128$):	0.013	0.018	0.024	0.025
$d = 2$ ($N = 16, 32, 48, 64$):	0.02	0.1	0.2	0.4

The two rows are measured at *different* configurations: $d = 1$ at $\lambda = 0.25$, $L = 32$, $\Delta E = 25$, $n = 8$; $d = 2$ at $\lambda = 0.1$, $L = 16$, $\Delta E = 10$, $n = 60$. The claim is the *dichotomy*, not a cross-row rate comparison.

The 1+1d rate is effectively marginal (its tadpole driver only logarithmic in d=1—slow, unbounded growth rather than a clean continuum limit; the rate 0.013 $\to$ 0.025 is still rising at N=128, though the growth over the final doubling is within the ensemble scatter at $n = 8$, so the quoted uncertainties are ensemble estimates)—while the 2+1d rate grows as $\sim k_{\max}^{2.1}$ with no saturation: a dimensional dichotomy in the pre-registered direction. (The top-cutoff d=2 rate scatters $\sim$20–40% across vacuum realizations—the d=2 rates are medians over 60 (the d=1 row over 8), quoted to one significant figure (d=1 to two), with the threshold checked on a $\approx$0.5 grid; the d=1 values likewise carry $\sim\pm$30–50% (upper-skewed) seed-to-seed scatter, and the fitted exponent carries a $\pm$0.2 ensemble uncertainty.)

But exponent decomposition (`qft_bridge_decomp.py`) reattributes the mechanism. Holding the source fixed (zero-point support truncated to $|k| \leq K_C$, N-independent up to $O((k\Delta x)^2)$ dispersion corrections ($\lesssim$3%)) and sweeping only the available channels, the purified channel factor is *flat* in both dimensions (fitted exponents -0.20 (d=2) and $+0.11$ (d=1); ensemble noise leaves room for a residual channel-log of at most $\sim$1.3$\times$ across the sweep—so the naive golden-rule channel prediction (linear in the cutoff, $\sim$4$\times$; a residual channel-*log* would give the weaker $\sim$2.2$\times$) is excluded a fortiori); holding the channels fixed and scaling the source amplitude a, the rate goes as $a^{4.96}$ (d=2) and $a^{4.45}$ (d=1) against the perturbative a^4; and the reconstruction reproduces the growth to within $\sim$30% in the exponent: the measured $\langle\phi^2\rangle_0 \propto k_{\max}^{1.16}$ raised to *half* the d=2 amplitude power ($4.96/2 = 2.48$, since the rate scales as $a^{4.96}$ while the variance scales as a^2) predicts $k_{\max}^{2.9}$, over-predicting the observed $\sim$2.1: the sign and rough size of the dimensional reversal are reproduced, the precise exponent not.

Claim 5.1 (the two ladders)**.** There are two dimension ladders. The *tadpole ladder*: the zero-point variance $\langle\phi^2\rangle = \int \mathrm{d}^d k/(2\omega(2\pi)^d) \sim k_{\max}^{d-1}$ (log at d=1, linear at d=2)—this is what classical-statistical lattice dynamics tracks, and it is the true mechanism of the marginal-versus-power-law dichotomy above. The *channel ladder*: the reduced decay measure of Prop. 3.1 plus the boost orbit—the CJM/GSTZ object. Classical dynamics cannot see it, for the standard structural reason: emission rates into a mode go as $(n+1)$ quantum mechanically but as n (stimulated only) in the classical wave limit—the missing spontaneous "+1" is exactly the golden-rule channel factor—so the classical cascade populates high k only by nonlinear local-in-k flux, insensitive to

the far cutoff. The cutoff pathologies of classical-statistical simulations seeded with zero-point noise are documented [41], including the cutoff-dependent $\langle\phi^2\rangle$ renormalization in real-time false-vacuum-decay lattices [31]; our decomposition quantifies them for the ghost problem.

Methodological consequence, stated plainly: classical-statistical vacuum simulations are tadpole-ladder probes; channel-ladder questions require quantum treatment (our channel toy above, the phase-space integrals of Prop. 3.1, or genuine quantum simulation). Our notebook documents the two artifacts caught en route (a vacuum-normalization error, retracted with scope; Verlet integrator drift at marginal timestep firing the $\lambda = 0$ control)—each alone would have produced a publishable-looking false trend, and each was caught by an inline control column.

6 The foliation

6.1 Beables anchor a slicing; ghosts make it visible

A beables-first field theory needs configurations, and field configurations live on *spatial slices*: choosing the configuration space is choosing a foliation. This is not peculiar to the present program—it is the known structure of Bohmian field theory [22], where the foliation may even be extracted covariantly from the quantum state itself [23] —and Barandes' own development stays on finite configuration spaces, deferring exactly this point [3]. Within the natural field-theoretic extension, then, the boost-orbit question of Sec. 4 is answered structurally: relative to the theory's own foliation, orbit members are distinct configurations—the multiplicity horn.

The standard objection to foliated ontologies is that the foliation is idle: in quantum equilibrium it is empirically invisible (no-signaling), pure metaphysical surplus. Our results replace "invisible" with a *criterion*:

Claim 6.1 (the equilibrium shield and its breaking)**.** *In quantum equilibrium*, the foliation is empirically hidden (the equilibrium shield): for Bohmian mechanics this *no-signaling* statement is Valentini's signal-locality theorem [37]. Its finite-configuration core in the present framework is Proposition 7.2, which assumes *neither* a bounded-below spectrum *nor* an equilibrium ensemble: marginal-independence holds for *any* dilated local unitary, with spectral positivity entering only in the interacting extension (Open Problem 7.3; cf. Sec. 7). One other equilibrium route to foliation observability has been proposed: Bohmian arrival-time statistics in EPRB configurations [45]—a laboratory-scale mechanism resting on the contested measurability of arrival times, disjoint from the ghost route argued here. A coupled ghost sector breaks the shield: the vacuum-decay observables of Sec. 4—the rate and the decay spectrum, isotropic precisely in the individuating frame (with a cutoff edge in that frame where a frame-cutoff rescue applies)—are *frame-dependent local observables* depending on the foliation (the trilemma's operative sense, Claim 6.2: some observable depends on the foliation), not a superluminal signaling channel. *Ghosts are foliation detectors.* (The no-signaling shield for the indivisible framework at full strength is a conjecture with a defined proof target, Open Problem 7.3; the breaking is Claim 4.3 plus kinematics.)

6.2 Determinacy: the foliation as a property of the state

The vacuum is Lorentz-invariant: $\langle T^{\mu\nu}\rangle \propto \eta^{\mu\nu}$ has no distinguished timelike eigenframe, so a DGNSZ-style extraction [23] returns nothing. We upgraded this algebra to a measured curve (`s3_foliation_determinacy.py`): ensembles combining a frame-symmetric zero-point part with boosted thermal occupations at temperature T, frame estimator $\hat{v} = P/E$ per realization. At

$T = 0$ the estimate is pure scatter (mean 0.001, scatter 0.12, SNR ≈ 0; at $T = 0.01$ the mean even lands on the wrong sign); the signal rises with the thermal share to SNR ≈ 51 at $T = 3$. *The foliation information content of a state vanishes continuously at the vacuum*—which is Theorem 4.1 in ontological dress: the state that determines no foliation is the state whose ghost dynamics is ill-posed (one direction only—a frame is necessary, not sufficient, to tame $q = \infty$). Conversely, any perfect-fluid-form state with $\rho + p \neq 0$ determines a frame—note the boundary, $\rho + p = 0$, is the phantom divide $w = -1$—so the actual cosmological state selects the comoving frame, and CJM's pragmatic choice of the CMB frame for the cutoff [7] is what a state-extracted-foliation theory *prescribes*.

6.3 The state proposes; the ultraviolet disposes

Is the state-selected frame dynamically stable? A lattice experiment (`foliation_m3.py`): both sectors prepared as thermal baths boosted to v relative to the lattice (traveling-wave construction; fixed spectral support, so the state is cutoff-independent), the ghost bath hotter; the frame of the secular energy transfer measured as $v^* = (\mathrm{d}P/\mathrm{d}t)/(\mathrm{d}E/\mathrm{d}t)$, which equals v for production isotropic in the state frame and 0 for lattice-anchored production. Result: a two-epoch structure. Deep in the evolution the transfer is UV-directed (at N=64, 54% of transferred energy lands in the band $3 < |k| \leq 6$, far above the bath support $|k| \leq 1.2$, and ∼83% above the support altogether; at N=128–256 over 90% lands above the support, much of it beyond $|k|$=6) and every band shows $v^* \approx 0$: the state's frame is erased en route to the ultraviolet. In the early window, continuum covariance *guarantees* state-frame anchoring as a matter of principle (the boosted evolution is the boost of the rest evolution, so $v^* = v$ exactly in the continuum limit); the lattice experiment, however, could not resolve it: small ensembles suggested a positive early-window signal, but a 100-run settlement showed no significant positive signal ($\lesssim 2\sigma$) at both N=128 and N=256 (screened means $+0.04 \pm 0.05$ and -0.03 ± 0.10; the slope-ratio estimator is heavy-tailed, and runs with negligible early-window transfer must be excluded or they dominate the raw mean). What the experiment establishes robustly is the erasure side—the UV-directed cascade and the absence of the state's frame in every band at late times—while providing no positive lattice evidence of the covariant window at accessible sizes. *The state proposes the foliation (by covariance); the ultraviolet disposes (by measurement).* This is the dynamical content of Garriga–Vilenkin's condition: their finite-age frame can hold only because their couplings are UV-soft [9]; in a UV-local theory the cascade hands the frame to the regulator.

6.4 The trilemma

Claim 6.2 (pick two)**.** Coupled ghosts; *empirical* Lorentz invariance (no *in-principle* observationally accessible preferred foliation—"accessible" meaning some observable depends on the foliation with nonzero effect, realized detectability being a separate, flux-dependent question; this is the operative sense, since a beables ontology always carries a foliation, hidden or not, by Sec. 6); definite configurations (beables): any two, never all three. The born-vacuum construction [9] sits on the first horn because its birth surface is *observable* through the frame-locked decay products, as are the cutoff frame [7], the condensate background [17], and explicit breaking [18].

- coupled ghosts + beables ⇒ the foliation becomes observable (Claim 6.1); every configuration-preserving rescue on record supplies and thereby exposes one;

- empirical Lorentz + beables ⇒ no coupled ghosts (healthy sectors; the ontological foliation exists but hides in quantum equilibrium);
- coupled ghosts + empirical Lorentz ⇒ no beables: the definition-level rescues—$\mathcal{PT}$/no-ghost quantization [19] (which the companion paper showed exits the sample space [1]) and Euclidean state-space surgery [20].

The claim is a statement about combinations, verified against every example we could find; it does not assert which pair nature chose. The incompatibility is a tree-level statement: were a resummed suppression to render the boost-orbit multiplicity sum convergent (a Dine–Draper–Park-style escape—GSTZ's open question, Sec. 9), the first horn would relax and a Lorentz-invariant coupled ghost with definite beables could survive. Two of its three *edges* are established prior art: ghosts demand a preferred frame ([7], and theorem-level for stable NEC violation [25]), and realist hidden-variable accounts of quantum correlations demand one (Hardy's theorem [43] and its descendants). The triangle—and the observation that the frame demanded by ghost regularization and the foliation demanded by beables can be *the same object*, jointly observable—is, to our knowledge, new.

Claim 6.3 (the divergence lives in the state)**.** At tree level, the particle-number expectation per unit volume $\langle N(t)\rangle/V$ of the evolving coupled-ghost vacuum carries the same [orbit volume] × [reduced integral] factorization as the exit rate (2): the number expectation sums the same final-state phase space as the golden-rule rate and boost-orbit members share a common particle number, so the orbit multiplicity factors out exactly as there. With the invariant reduced integral held ultraviolet-finite and nonvanishing (its ordinary UV divergence regulated as usual), it diverges iff the multiplicity sum does—so the genuinely instantaneous divergence is the boost-orbit one, an ontology-neutral expectation invoking no beable configuration space, though it remains conditional on the tree-level state. (Channel-toy verification, `s5_state_divergence.py`: $\mathrm{d}\langle N\rangle/\mathrm{d}t$ tracks $M\Gamma_1$ within 3% to $M = 32$ for the flat orbit measure, versus a plateau for the decaying measure.)

Remark 6.4 (event ontologies inherit the dichotomy). "Definite configurations" above means configurations on slices; the known foliation-free alternatives—event ("flash") and mass-density ontologies, as in relativistic GRW [57, 28] (whose interacting extension is delicate; cf. the no-go of [27])—have rates that are *functionals of local quantum-state content*. By Claim 6.3, they do not escape the trilemma; they import its central question: on the multiplicity reading the state acquires unbounded particle content and any content-proportional event rate diverges with it—instantaneity in event clothing—while a covariant flash law that takes the redundancy reading must dispute the tree-level state itself, which is GSTZ's open question once more. The global-existence (non-explosion) theorems for Bell-type QFT jump processes [46] make the point citable: their hypotheses on the Hamiltonian are exactly what a ghost sector violates. The flank thus narrows from "event ontologies might escape" to "event ontologies face the same multiplicity-or-redundancy fork," with the resummation question open for all parties equally.

7 Causality and signaling

Where, if anywhere, can superluminal signaling enter this framework? We identify two entry points, and both were simulated.

7.1 Conditioning: the Lee–Hsieh–Flammia–Lee signal, located and quantified

Ref. [21] showed that local $\mathcal{PT}$ evolution on one arm of an entangled pair, treated as fundamental, permits superluminal signaling. The controversy has a resolution literature we build on: the apparent signal was experimentally located in the post-selected subspace [38], the metric-corrected treatment restores consistency [39], and the no-go theorems survive proper embedding [40]—our treatments (iii), (i), and (ii) respectively. What the stochastic currency adds is all three in one classical-probability bookkeeping, plus the closed form below. In this currency (`s1_no_signaling.py`; Bell pair, Alice's message the loss orientation of a dimer, signal the total-variation distance between Bob's marginals): (i) metric-corrected closed dynamics, max TV 1.7×10^{-16}; (ii) dilated open dynamics without post-selection, max TV 4.4×10^{-16}—the framework's falsifiable core, clean; (iii) survival-conditioned dynamics: the signal appears, with closed-form magnitude.

Theorem 7.1 (the conditioning signal, in closed form)**.** *For the Bell state and the broken-phase dimer of [1] (loss rate* $\sin\theta$*, coupling* s*—the companion's dimer notation, unrelated to the invariant mass* s *of Secs. 3–4—*$s < \sin\theta$*, so* $\tilde{\kappa} = \sqrt{\sin^2\theta - s^2}$ *is real), with* $p = c + \sin\theta\,\sigma$*,* $q = c - \sin\theta\,\sigma$*,* $r = s\sigma$ *(*$c = \cosh\tilde{\kappa}t$*,* $\sigma = \sinh(\tilde{\kappa}t)/\tilde{\kappa}$*; this dimer* σ *is unrelated to the residual measure* $\sigma_d(s)$*, and the amplitude* q *here to the exit rate* q *of the explosion dictionary above), the survival-conditioned signal between the two loss orientations is*

$$\mathrm{TV}(t) = \frac{|p^2 - q^2|}{p^2 + q^2 + 2r^2}, \qquad \mathrm{TV}(t \to \infty) = \frac{\tilde{\kappa}}{\sin\theta} = \sqrt{1 - \left(\frac{s}{\sin\theta}\right)^2}\,: \tag{3}$$

zero at the exceptional point, approaching unity deep in the broken phase.

Proof. $(C \otimes \mathbf{1})|\psi_{\mathrm{Bell}}\rangle$ gives Bob's conditioned marginal $p_B(b) = \|C|b\rangle\|^2 \,/\, \sum_i \|C|i\rangle\|^2$; the squared column norms of the broken-phase contraction are $\mathrm{e}^{-2\sin\theta\, t}(p^2 + r^2)$ and $\mathrm{e}^{-2\sin\theta\, t}(q^2 + r^2)$ by the closed forms of [1]; the two messages exchange $p \leftrightarrow q$. The asymptote follows from $c, \tilde{\kappa}\sigma \to \frac{1}{2}\mathrm{e}^{\tilde{\kappa}t}$ and $\tilde{\kappa}^2 + \sin^2\theta + s^2 = 2\sin^2\theta$. Numerics: (3) matches to 2.2×10^{-16}; the s-sweep matches $\tilde{\kappa}/\sin\theta$ to six digits away from the exceptional point. □

(The unbroken-phase construction of [21] itself is treatment (iv) in the script: conditioned signal up to 0.577, identically zero under treatment (i) with the same Hamiltonians.) The interpretation is exact: the "FTL" of local $\mathcal{PT}$ lives only in treating the survival-conditioned (sub-stochastic, renormalized) process as the whole sample space—and even there it is not operational, since constructing the conditioned ensemble requires Alice-side survival information that travels classically. Same Hamiltonians, three bookkeepings, signal in exactly one.

7.2 Completions: messaging through infinity

The companion paper showed conservative completions of an explosive process return probability from infinity after the finite classical round trip [1]. We upgraded the observation to a communication protocol (`s2_infinity_channel.py`): Alice gates the right exit of $V = -x^4$ (explosive) or $V = -x^2$ (marginal) with a barrier (her bit); Bob is a direction-resolved current detector near the left boundary, triggering only on flux entering from infinity; the completions are periodic identification (conservative "wormhole") versus absorbing layers (minimal process). All twenty-four runs (two potentials × two completions × three separations $L = 8, 10, 12$ × Alice's two bit settings) are clean (the $L = 6$ geometry is degenerate—Alice's station overlaps the launch packet—and is excluded, as recorded in the notebook):

	conservative (wormhole)	minimal (absorbing)
$V = -x^4$ (explosive)	one-bit channel; latency 0.18/0.15/0.16 at $L = 8/10/12$: **distance-independent**	silence ($J_{\max} \leq 3 \times 10^{-5}$); norm $\to 0.000$
$V = -x^2$ (marginal)	channel exists; latency 0.94/1.02/1.10: **grows with** L	silence; *in-box* norm $\to 0.000$

The marginal potential $V = -x^2$ is essentially self-adjoint (limit-point at infinity), so its true minimal process is *conservative*; the "norm $\to 0.000$" in that row is the in-box (finite-window) norm as probability leaves the simulation box, and its wormhole/absorbing variants are box-boundary *controls*, not genuine self-adjoint completions. Only $V = -x^4$ is limit-circle, with real self-adjoint-extension freedom—so the three-completion taxonomy below is a statement about $-x^4$.

A third completion completes the taxonomy (`s2b_dirichlet.py`): hard Dirichlet walls fire the *right-side* direction-resolved detector (reflection off infinity, $t_B \approx 0.23$, distance-independent, left detector silent, norm conserved)—so the three completions of one interior Hamiltonian (the explosive $V = -x^4$) produce three distinct signatures: return from the left (wormhole), return from the right (walls), silence (minimal). The same interior Hamiltonian thus contains a distance-independent messaging channel through infinity, a distance-penalized one, or none at all, according to the *completion*—regulator data, not physics in H. That extension choices are experimentally distinguishable has precedent at the *origin*-singularity anomaly (graphene supercritical Coulomb ladders [29]); the taxonomy here is its counterpart at infinity. The model is nonrelativistic—there is no light cone, and "acausal" here means latency that fails to scale with separation, the model's stand-in for light-cone violation. In that operational sense, the acausal channel occurs—in the cases tested ($-x^4$ and $-x^2$), and as the time-of-flight dictionary predicts—exactly in the explosive class (the dictionary's time-of-flight boundary, visible as a flat versus rising latency curve): *causality selects the minimal process.* For a malicious ghost, probability conservation and causality cannot both be kept in the completions tested.

7.3 Synthesis: positivity as the pillar

Assembling Secs. 6 and 7: within a beables ontology, apparent superluminal signaling enters at the two points we identify (the conditioning point provably the unique marginal-breaker in the finite core, over the framework's dilated local dynamics (Prop. 7.2); the completions door and field-theoretic exhaustiveness remain open)—conditioning fallacies and acausal completions. The acausal-completions door and the equilibrium shield are gated by *spectral positivity* (bounded-below sectors give conservative processes and no channel; a ghost sector opens them at once); the conditioning door, by contrast, is a post-selection artifact present even in bounded-below lossy systems (Theorem 7.1), gated by whether one conditions on survival rather than by positivity. The finite-configuration core of the shield—no-signaling for *any* dilated local dynamics, positivity entering only in the interacting extension (Open Problem 7.3)—is a theorem:

Proposition 7.2 (no-signaling core of the shield)**.** *Let the joint configuration space be $C_A \times C_B$ (finite), let Alice's operation be any dilated local dynamics given by a* unitary U_A *on $A \otimes \mathrm{anc}_A$, Bob's any* unitary U_B *on $B \otimes \mathrm{anc}_B$, and let the joint process be $W = |U_A \otimes U_B|^2$ from an arbitrary joint amplitude Ψ, extended by the fixed initial ancilla configurations, with Bob's marginal summed*

over his ancilla configurations. Then Bob's marginal at every time is independent of Alice's choice of U_A.

Proof. $p_B(b) = \sum_{a',\beta} |\langle a', b, \beta|(U_A \otimes U_B)\Psi\rangle|^2 = \langle\Psi|(U_A^\dagger U_A) \otimes \big(U_B^\dagger(|b\rangle\langle b| \otimes \mathbf{1}_{\mathrm{anc}_B})U_B\big)|\Psi\rangle = \langle\Psi|\mathbf{1} \otimes U_B^\dagger(|b\rangle\langle b| \otimes \mathbf{1}_{\mathrm{anc}_B})U_B|\Psi\rangle$, with a' running over Alice's configurations *including* ancilla (leak) configurations and β over Bob's ancilla. Survival conditioning breaks exactly the first equality (a projector is inserted on the A side), which is where—and only where—Theorem 7.1's signal lives. □

The closed-unitary case of Proposition 7.2 is the no-signaling reading of the causal-locality condition Barandes, Hasan, and Kagan impose [26] (whose marginal-factorization equations give it directly); our statement extends it to dilated (open, lossy) local dynamics and identifies survival conditioning as the unique breaking point.

What remains open is the field-theoretic and statistical extension:

Open Problem 7.3 (the shield theorem, full strength)**.** Extend Proposition 7.2 to field configurations with interacting joint dynamics and quantum-equilibrium ensembles: prove that spectral positivity of all interacting sectors implies no-signaling with respect to the foliation (invisibility of the slicing), and that the implication is sharp—i.e., any coupled sector unbounded below yields foliation-dependent observables (the converse direction of Claim 6.1).

8 Observational and experimental probes

The trilemma is empirically loaded through its conjunctions.

(a) Dark energy. Robust confirmation of $w < -1$ would make the ghost hypothesis live and activate the rest of this section; $w \geq -1$ retires it as the dark-energy explanation. Currently DESI DR2 prefers evolving dark energy over ΛCDM at 2.8–4.2σ [32] (see [33] for a review)—with the honest caveat that the quoted significance depends on which supernova sample is combined, a principled combination giving $\sim$3.1σ [34].

(b) The MeV gamma-ray background, with a frame signature. On the multiplicity horn, a coupled phantom fills the sky with photons up to the cutoff, already bounded by the diffuse gamma-ray data [7, 9, 48]; the current bound is $\Lambda \lesssim 19\,\mathrm{MeV}$ [48], which places the spectral edge of a bound-saturating phantom inside the unobserved "MeV gap." The stochastic reading adds the *foliation signature*: the background is isotropic in the individuating frame, so a detector moving at $\beta = v/c = 1.23 \times 10^{-3}$ sees a Compton–Getting dipole [59] aligned with the CMB dipole (the kinematic-dipole toolkit for cosmological backgrounds is standard [44], including threshold-dependent amplitudes [30]) *and* a direction-modulated spectral edge $\Lambda(\theta) \simeq \Lambda(1 + \beta\cos\theta)$. A mock-sky pipeline (`s4c_dipole_template.py`; exact aberration and Doppler transport, estimators verified against their analytic amplitudes) gives the template: count-rate dipole amplitudes 3.7–4.3×10^{-3} depending on the threshold observable (all-photon rate 3β; thresholded counts add $\beta E_{\mathrm{thr}}/(\Lambda - E_{\mathrm{thr}})$; each mock photon carries the arrival-rate weight $1 + \beta\cos\theta$—an unweighted mock measures only the aberration part 2β and is blind to the rate factor, an error caught in our own audit and documented in the notebook), requiring **4.0–5.5×10^6 signal photons for** 5σ with the apex fixed by the CMB dipole; and the edge modulation (0.12%)—whose cosmologically integrated arriving spectrum is a linear *wedge* rather than a sharp step (`s4d_cosi_forecast.py`), costing ×5–10 in counts—is recoverable *through* realistic 3% instrument resolution by percentile regression at $\sim 3\sigma$ with a few $\times 10^6$ photons (5σ near 10^7). Dilution: for signal fraction f_s

of collected diffuse counts, N scales as $1/f_s^2$—and CJM's bound is set precisely by the signal *saturating* the observed diffuse background, so *at* the bound $f_s = O(1)$ and the measurement is exposure-limited. This is a knife-edge, not a generic expectation: $\Gamma \propto \Lambda^8$ means a cutoff at half the bound gives $f_s \sim 4 \times 10^{-3}$ and inflates N by $\sim 6 \times 10^4$. Two further practical caveats: the measured GeV-band diffuse dipole is percent-level and dominated by non-kinematic (astrophysical) components [42], so extracting a 10^{-3} kinematic dipole requires exposure-uniformity control at that level; the edge observable is less exposed to both issues. The discriminant against mundane comoving backgrounds (which share the Compton–Getting dipole) is the edge modulation together with the absence of source correlation.

(c) Short-range gravity. Garriga and Vilenkin already note that their cascade bound ($\mu \sim 10^{-3}\,\mathrm{eV}$) is only marginally consistent with 0.1 mm tests of the inverse-square law [9, 10], leaving a narrow surviving Lorentz-invariant window near $\mu \approx 2 \times 10^{-3}\,\mathrm{eV}$; improved torsion-balance experiments can close it, eliminating the born-vacuum rescue outright.

The decision matrix. The decay signal is a *joint* observable: vacuum decay requires genuinely negative-energy states (a modified-gravity mimic of $w < -1$ produces no such background), and the signal's frame-locking *is* the foliation. One positive measurement therefore confirms both hypotheses at once, and the residual degeneracies close with one added axis—structure-growth and perturbation data distinguishing a genuine phantom field from a gravitational mimic:

	signal seen (frame-locked dipole + edge)	**signal excluded** (window closed)
phantom field established ($w < -1$ + growth data)	ghosts *and* foliation both confirmed, in one measurement	ghost real but no *observable* decay: evidence *against* definite configurations *if* the frame cutoff saturates the diffuse-background bound—then the exit-the-sample-space horn [19, 20] wins (at tree level)
phantom excluded	(inconsistent under the phantom-DE identification; a signal would otherwise indicate a non-DE ghost sector)	the observable ghost hypothesis retired; foliation undecided

Both axes of the matrix are simulation-backed. The rows (`s4a_books_balance.py`): identical lattices, identical temperatures, identical coupling strength, only the partner sector's energy sign flipped—the healthy pair *equilibrates* (transfer saturates at 7.4 ± 1.0, the equipartition scale, for the entire second half of the run), while the ghost pair runs away without bound (past 10^3 by $t \approx 120$; no joint equilibrium exists [8]): only a genuine phantom feeds an unending flux. The columns (`s4b_horn3_silence.py`): one two-mode model, three quantizations—the ghost quantization reproduces the exact decay law $\langle n \rangle = \sinh^2(gt)$ to five decimals through $gt \lesssim 1.2$ (truncation tail thereafter); the positive-spectrum requantization gives *machine-exact zero* at every time (the vacuum is an eigenstate); the healthy control shows only bounded virtual pairs. A decay signal exists in exactly one quantization.

Three of the four cells are informative—one (phantom established, signal seen) decisive for both questions, a second (phantom established, signal excluded) decisive for both only where

the predicted flux is detectable (see below), the third (phantom excluded, signal excluded) for the ghost question alone; the fourth is inconsistent under the phantom-DE identification. The signal-excluded verdict *against* beables is decisive only where the predicted flux is detectable: a frame-cutoff phantom whose cutoff Λ sits well below the diffuse-background bound decays unobservably slowly with its configurations intact, and $\Gamma \propto \Lambda^8$ makes this *undetectable*-Λ region wide (the *detectable* band above being the knife-edge of the dilution estimate), leaving that cell undecided for the foliation question. Conditional on the top row—exactly the measurement DESI-era surveys are making—the gamma-ray program is a *two-sided* test of the foliation: the surviving window is already pinched to $\mu \approx 2 \times 10^{-3}$ eV between the cascade bound and short-range gravity [9], so "window closed" is a realistic outcome, and it would count against the born-vacuum rescue outright (the low-Λ frame-cutoff region remaining undecided, as above). The bottom-right cell states the honest floor: in a ghost-free equilibrium universe the foliation is undecidable *in principle*, because the equilibrium shield is quantum mechanics' own no-signaling property—an experiment exposing a hidden foliation there would contradict quantum predictions themselves. (One speculative shield-breaker in the literature is primordial quantum *non*-equilibrium [58], which would expose the foliation without ghosts; the contested Bohmian arrival-time route (Sec. 6) [45] is another. We note both and set them aside.) Given ghosts, nature's choice among the trilemma's horns is a measurement program, not a philosophy seminar.

(d) Analog platforms. The ladder criterion (Lemma 2.1) is realizable in cavity/circuit QED: two-mode squeezing already is the transient cascade ($\langle n \rangle = \sinh^2 gt$ is routinely measured), and engineered superlinearly Bose-enhanced couplings would exhibit the transition toward finite-time runaway at $\alpha_{\text{eff}} > 1$. The frame-competition experiment of Sec. 6 (state frame versus medium frame of an instability's emission) is native territory for polariton fluids and analog-gravity systems. The companion paper's photonic tests (the terminal-indivisibility closed forms in lossy couplers) probe the quantum-mechanical layer of the same program [1].

9 Discussion: what is established, and at what strength

Theorem-grade: the Fock ladder criterion and growth-law lemma; instantaneity on the multiplicity horn (Thm. 4.1); the conditioning-signal closed form (Thm. 7.1); the no-signaling core (Prop. 7.2). *Measured*: the factorization verification and the dimension ladder (Prop. 3.1); the completions table; the tadpole/channel decomposition; the two-epoch frame competition; the determinacy curve. *Argued*: the final-state-invariance criterion's verdict on tree-level ghost decay; frame individuation; the state-content factorization (Claim 6.3); the trilemma (a verified pattern over known examples). *Conjectured*: the shield theorem at full strength (Open Problem 7.3; its finite core is proved).

Limits, stated without decoration. Everything golden-rule–based inherits perturbation theory (Remark 4.4); the degeneracy of orbit members is a tree-level statement, and a resummed suppression would render the multiplicity sum convergent (a Dine–Draper–Park-style escape, distinct from redundancy)—that is GSTZ's open question, not ours to close. The lattice results are classical-statistical, with the stated blindness to spontaneous emission; the factorization is numerically verified to 0.1% in $d = 1$ and to ∼3–7% by Monte Carlo in $d = 3$ (Prop. 3.1), still higher d resting on the invariance argument and GSTZ's analytics. The foliation claims live within the natural field-theoretic extension of the beables framework, which its author has not committed to; and no experiment distinguishes the ontology itself—only its conjunction with ghosts is testable. Finally, the trilemma does not say which two nature picked; it says the three

advertised virtues of fundamental physics cannot all be kept, at tree level, if ghosts exist, and it prices each exit.

Adjacent results, positioned. The active ghost-stabilization program of Deffayet et al. proves benign behavior for finite-dof quantum mechanics [50] and demonstrates it numerically for classical 1+1d fields [49]; neither engages the boost orbit, and both are consistent with our taxonomy (the QM constructions live in the companion's benign/transient classes; the 1+1d stability sits at the mild end matched by our marginal (logarithmic) d=1 rate). Buoninfante's analysis [54]—ghosts above threshold admit no asymptotic particle interpretation—is the sharpest operator-theoretic challenge to any "ghost decay" framing; our object is the vacuum's exit rate on a configuration sample space rather than an S-matrix ghost width, but if the dressed vacuum ultimately supports no decay interpretation, that would favor the convergent readings and we flag it. Könnig et al.'s slow heavy-ghost decay [52] and Kaplan–Sundrum's energy-parity construction [56] are horn-one examples (frame-anchored cutoffs); Babichev's identification of ghost decay with Cherenkov radiation [53] makes the frame-anchoring vivid; the modern Γ/V formalism is [55]; and the information-flow criticality of $\mathcal{PT}$ systems [51] is the quantum-state-level cousin of the conditioning analysis of Sec. 7.

The companion paper ended by observing that the ghost problem, in the stochastic currency, is a recurrence-classification problem. The field-theoretic sequel is that it is, more deeply, a *sample-space* problem: the divergent vacuum rate, the regulator dependence, the foliation, and every apparent causality violation we examined are one question—*which configurations are distinct, and relative to what*—asked at increasing energy. Every pathology entered through that door, and every resolution on record pays at that door.

Author contributions and use of AI

This manuscript was produced through a collaboration, directed by the author, with the large language model Claude Fable 5 (Anthropic). The AI system carried out essentially all of the technical work: the research program and its strategy, the derivations and proofs, the design, execution, and interpretation of all simulations, the literature analysis, and the drafting of the manuscript, including this statement. J.T. posed the research directions, provided iterative review and quality pressure, and takes sole responsibility for the decision to circulate the manuscript. In line with prevailing editorial policies, the AI system is not listed as an author. The research notebook in the public repository records the work hour-by-hour, including three same-day corrections of our own errors and one retraction, so that the process as well as the results can be audited.

Code availability

This paper's scripts, source, and notebook excerpt are public at

`https://github.com/embeddetech/p03-instantaneous-ghosts`

The full running lab notebook (`notes.md`), the companion papers, and the provenance record live in the program's hub repository, `https://github.com/embeddetech/ghosts-program`. This paper's results are reproducible from the following scripts, each standalone (NumPy/SciPy), running in seconds to minutes on a laptop, and printing its controls inline:

- `qft_bridge_stage2.py` — ladder onset, growth laws, channel multiplicity
- `qft_bridge_stage3.py` — factorization, dimension ladder
- `qft_bridge_e4_2d.py` — the dimensional dichotomy
- `qft_bridge_decomp.py` — tadpole/channel decomposition
- `foliation_m3.py` — the frame competition
- `s1_no_signaling.py`, `s2_infinity_channel.py`, `s3_foliation_determinacy.py` — the signaling, completion, and determinacy experiments
- `s4a_books_balance.py`, `s4b_horn3_silence.py`, `s4c_dipole_template.py`, `s4d_cosi_forecast.py` — the decision-matrix discriminants, the detection template, and the diffuse-background / COSI-era detectability forecast
- `s5_state_divergence.py` — the state-content observable of Claim 6.3
- `qft_bridge_closeout.py` — $d = 3$ factorization Monte Carlo; Garriga–Vilenkin log-growth replication
- `s2b_dirichlet.py` — the third completion (Dirichlet walls) of the infinity-channel taxonomy

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